has a thick stem which does most of the food production through photosynthesis and leaves are reduced to small spines with less surface area.

b. Sub – aerial Modifications

Stem of some plants remains sub – aerial which grow horizontally on the surface of the soil for the purpose of reproduction. There are four types.

Runner

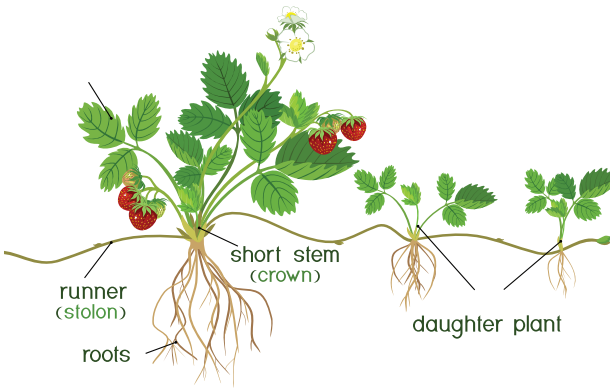
The stem which grows laterally on the surface of the soil, breaks up to produce roots where it touches the ground to give rise **to new plants. E.g. Centella (Vallarai)**



Centella

Stolon

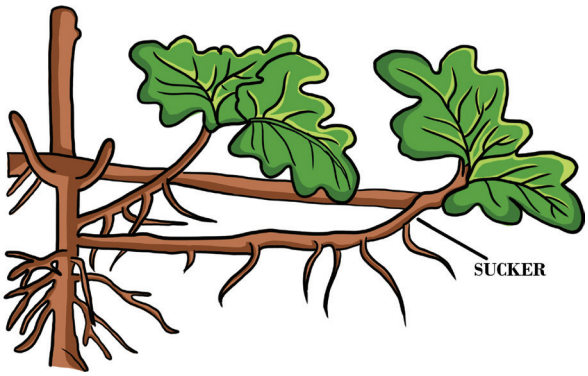
Stolon is a slender branch of the stem that grows upwards to some distance and then bends towards the ground. Upon touching the ground, it gives rise to a new plant. **E.g. Wild strawberry**



Wild strawberry

Sucker

Sucker is a short and weak lateral branch that grows diagonally upwards and directly gives rise to a new shoot. **E.g. Chrysanthemum**



Chrysanthemum

Offset

An offset is a short and thick branch that arises from the axial part of a leaf. It has thick internodes. It produces a tuft of leaves and cluster of small roots below. **E.g. Eichhornia**



Eichhornia

c. Underground modifications

In aerial and sub aerial modifications, stem has indefinite growth. In underground modified stem, whole stem is burried under the ground and it has definite growth. Usually stem grows above the ground, but there are some stems that grow under the ground to store food. These underground stems swell and become thick. There are four types of underground stems. They are:

1. Rhizome 2. Corm
3. Tuber 4. Bulb